

Abhinand Meethele Valappil

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Summary

Computer Science Engineering student at SRM Institute of Science and Technology (CGPA: 9.20/10, graduating May 2027) with hands-on experience developing full-stack web applications, distributed systems, and AI/ML-powered solutions. Experienced in designing scalable, service-oriented applications and integrating intelligent capabilities into production-style workflows. Strong foundation in software engineering, system design, and application development, with a focus on building reliable and practical solutions. First author of a research paper published in IEEE Xplore.

Technical Skills

Languages: Python, C/C++, SQL

Backend & Distributed Systems: Node.js, Express.js, FastAPI, Flask, RabbitMQ, Redis, Socket.io, REST APIs, Docker

Frontend: React.js, Bootstrap, EJS, HTML, CSS, JavaScript

Databases: PostgreSQL, MongoDB, SQLite

AI/ML Integration: Hugging Face Transformers, LangChain, Scikit-learn, TensorFlow, Pandas, NumPy

Cloud & Tools: Google Cloud, Firebase, Vercel, Git, GitHub, Claude Code CLI

Projects

Microservices Chat Application

Mar 2026

Node.js, TypeScript, RabbitMQ, Redis, Socket.io, MongoDB, Next.js | GitHub

- Designed a service-oriented architecture with three independently deployable services (Chat, User, Mail) communicating asynchronously over RabbitMQ, decoupling OTP delivery from the request path so authentication is never blocked by mail latency.
- Implemented real-time messaging, typing indicators, and presence over Socket.io, backed by Redis for session storage and pub/sub fan-out across service instances.

ColabSphere

2026 – Present

Node.js, Express, React, Google Cloud Platform, Firebase | Live

- Building a collaboration platform that matches developers to open projects, with a Node/Express REST backend deployed on Google Cloud Platform and a Firebase-hosted React client.
- Implementing authentication, project-contributor matching, and persistence against Firebase services.

AI Clipper & Caption Generator

Jul 2025 – Nov 2025

Python, Next.js, React, OpenAI Whisper, Gemini API | GitHub

- Built an end-to-end pipeline that transcribes long-form video with OpenAI Whisper, segments it, and passes candidate spans to the Google Gemini API to detect highlight-worthy moments.
- Generated platform-ready captions from the detected segments and exposed the workflow through a Next.js interface.

Research & Publications

Enhanced Compression Model (ECM): A Hyperprior-Based Neural Image Compression Framework

Jul 2026

1st Author | Published, IEEE Xplore: [Link] | presented at INSTCon 2026, NIT Rourkela

- Extended a Scale Hyperprior VAE architecture with semantic gating, channel attention, autoregressive context modelling, an entropy parameter network, and scale refinement; achieved 0.176 BPP, 30.882 dB PSNR, and 0.985 MS-SSIM on the Kodak benchmark, outperforming the Scale Hyperprior baseline and classical codecs (JPEG, WebP) at comparable or lower bitrates.

Education

SRM Institute of Science and Technology, Delhi NCR

Aug 2023 – May 2027

B.Tech., Computer Science and Engineering — CGPA: 9.20 / 10

St. Francis School, Indirapuram

2021 – 2023

CBSE Class XII (PCM): 82.8% | CBSE Class X: 88.5%

Certifications & Competitions

- Natural Language Processing – NPTEL (May 2026) [Link] | Foundations of Coding Full-Stack – Microsoft (Jan 2026) [Link] | EDA for ML – IBM (Feb 2026) [Link] | Python (Basic) – HackerRank (Oct 2024) [Link]
- Finalist – India AI Impact Buildathon, HCL GUVI (Feb 2026) [Link] | Participant – Adobe India Hackathon [Link], Amazon ML Challenge 2025, L'Oréal Sustainability Challenge 2025 [Link], Guidewire DevTrails

Languages

English (Professional) | Hindi (Native) | Malayalam (Native) | German (Elementary)